

Group Theory 3

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Claim 1A. Let $f: G \rightarrow H$ and $q: G \rightarrow Q$ be homomorphisms of groups. Show that f factors through q (via a homo $Q \rightarrow H$) iff $\text{Ker}(q)$ is contained in $\text{Ker}(f)$.

Proof. ($\implies$): Suppose f factors through q , via a homomorphism $F: Q \rightarrow H$ s.t $f = F \circ q$. For all $x \in \text{ker } q$, by definition of the kernel of a homomorphism $q(x) = e_Q$. Furthermore by definition of a homomorphism, $F(e_Q) = e_H$. Thus $f(x) = (F \circ q)(x) = F(q(x)) = F(e_Q) = e_H$. Thus by definition of kernel, $x \in \text{ker}(f)$. Thus every element in the kernel of q is also in the kernel of f , thus $\text{ker}(q) \subseteq \text{ker}(f)$.

($\impliedby$): Suppose $\text{ker}(q) \subseteq \text{ker}(f)$. For all $a, b \in G$ s.t $q(a) = q(b)$, $q(a)q(b)^{-1} = e_Q$. Since homomorphisms keep group structure and take inverses to inverses by definition, $q(a)q(b)^{-1} = q(a)q(b^{-1})$. Furthermore by definition of groups, $q(ab^{-1}) = e_Q$. Thus by definition of kernels we know, $ab^{-1} \in \text{ker}(q)$. By our assumption, $\text{ker}(q) \subseteq \text{ker}(f)$, thus $ab^{-1} \in \text{ker}(f)$. Thus $f(ab^{-1}) = e_H$. By definition of homomorphisms, $f(ab^{-1}) = f(a)f(b^{-1}) = f(a)f(b)^{-1} = e_H$. Thus $f(a) = f(b)$. Thus $q(a) = q(b) \implies f(a) = f(b)$. By the prehomework, f factors through q . q must be surjective for f to factor through q with a homomorphism $\square$

From now on assume q is surjective.

Claim 1B. If f factors as a function, it factors as a homomorphism (ie if we can find a function $Q \rightarrow H$ of the sort we want, we can find a group homomorphism).

Proof. Let $f: G \rightarrow H$ and $q: G \rightarrow Q$ be homomorphisms of groups. Suppose f factors through q as a function. Let F be the function s.t $f = F \circ q$.

For all $g \in G$, $f(g) = F(q(g))$. If both $x, y \in q_*(G)$ which by definition of surjective functions is true: By definition of a image there exists a $a, b \in G$ s.t $q(a) = x$ and $q(b) = y$. Furthermore since $f = F \circ q$ and $f(a) = F(q(a)) = F(x)$ and $f(b) = F(q(b)) = F(y)$. Thus $f(a)f(b) = F(x)F(y)$ and $f(ab) = F(q(ab)) = F(q(a)q(b)) = F(xy)$. Since f is a homomorphism, $f(a)f(b) = F(x)F(y) = F(xy) = f(ab)$. Thus F is a homomorphism. $\square$

Claim 2A. Show that if H is an Abelian group, and G a group, then $\text{Hom}_{\text{groups}}(H, G)$ is naturally a group.

Proof. Let H be a Abelian group, and G a abelian group. Let $A = \text{Hom}_{\text{groups}}(H, G)$ with binary operation $*$ s.t $f, g \in A$ $(f * g)(x) = f(x)g(x)$.

1. **Closure:** For all $x, y \in H$ and $f, g \in A$. $(f * g)(xy) = f(xy)g(xy) = f(x)f(y)g(x)g(y)$. Because G is abelian, $f(x)g(x)f(y)g(y) = (f * g)(x)(f * g)(y)$. Thus A is closed under $*$.

2. **Identity:** $\exists f \in A$, s.t $\forall h \in H$, $f(h) = e_G$. For all $g \in A$, $\forall h \in H$, $(g * f)(h) = g(h)f(h) = g(h)e_G = g(h)$ and $(f * g)(h) = f(h)g(h) = e_G g(h) = g(h)$. Thus f is the Identity element of A .
3. **Inverse:** $\forall f \in A$, $\exists g \in A$ s.t $\forall h \in H$, $g(h) = f(h)^{-1}$. $\forall h \in H$, $(f * g)(h) = f(h)g(h) = f(h)f(h)^{-1} = e_H$ and $(g * f)(h) = g(h)f(h) = f(h)^{-1}f(h) = e_H$. Thus every element has a inverse.
4. **Associativity:** $\forall a, b, c \in A$, $\forall h ((a * b) * c)(h) = (a * b)(h)c(h) = a(h)b(h)c(h)$ and $(a * (b * c))(h) = a(h)(b * c)(h) = a(h)b(h)c(h)$. Thus $((a * b) * c)(h) = (a * (b * c))(h)$ and A is associative.

Thus A is a group. $\square$

Claim 2B. Show that $\text{Hom}_{\text{groups}}(\mathbb{Z}, G)$ is naturally iso to G (where Z means integers under $+$).

Proof. Let $f : \text{Hom}_{\text{groups}}(\mathbb{Z}, G) \rightarrow G$ s.t it maps $\phi \rightarrow \phi(1)$.

Suppose $f(\phi_1) = f(\phi_2)$. Then, $\phi_1(1) = \phi_2(1)$. Since ϕ_1 and ϕ_2 are group homomorphisms acting on the group of integers with addition, thus $\phi(n) = \phi(1^n) = \phi(1)^n$ (not exponentiation is in context of group exponentiation), they are completely determined by their values on 1. Thus, $\phi_1 = \phi_2$, and ϕ is injective.

For any $g \in G$, consider the homomorphism $\phi_g : \mathbb{Z} \rightarrow G$ defined as $\phi_g(n) = g^n$. For all $a, b \in \mathbb{Z}$ $\phi_g(ab) = g^{ab} = g^a g^b = \phi_g(a)\phi_g(b)$. Now, notice that $f(\phi_g) = \phi_g(1) = g^1 = g$. Since we can find such a homomorphism ϕ_g for any $g \in G$, f is surjective.

Since f is surjective and injective, it is bijective. Also since its also a homomorphism it is a isomorphism. $\square$

Claim 2C. Show that $\text{Hom}_{\text{groups}}(\mathbb{Z}/n\mathbb{Z}, G)$ is naturally a subgroup of $\text{Hom}_{\text{groups}}(\mathbb{Z}, G)$. Which subgroup?

Proof. Let $\text{Hom}_{\text{groups}}(\mathbb{Z}/n\mathbb{Z}, G) = B$ and $\text{Hom}_{\text{groups}}(\mathbb{Z}, G) = A$. Define $f : B \rightarrow A$ s.t for all $\phi \in B$ and $\forall x \in \mathbb{Z}$, $f(\phi)(x) = \phi([x]_n)$. For all $\phi, \theta \in B$ s.t $f(\phi) = f(\theta)$, then for all $x \in \mathbb{Z}$ $f(\phi)(x) = \phi([x]_n) = \theta([x]_n) = f(\theta)(x)$. This means for all values of $[x]_n \in \mathbb{Z}/n\mathbb{Z}$, $\phi([x]_n) = \theta([x]_n)$. Thus $\phi = \theta$. Thus f is injective. Furthermore f is by definition surjective into its image. f is isomorphism into its image. Since $f_*(B) \subset A$ then B is isomorphic to a subset of A .

Furthermore since $\mathbb{Z}/n\mathbb{Z}$ and G are abelian then $\text{Hom}_{\text{groups}}(\mathbb{Z}/n\mathbb{Z}, G)$ is a group. Thus $\text{Hom}_{\text{groups}}(\mathbb{Z}/n\mathbb{Z}, G)$ is a subgroup of $\text{Hom}_{\text{groups}}(\mathbb{Z}, G)$.

Furthermore since we know $\text{Hom}_{\text{groups}}(\mathbb{Z}, G)$ is iso to G . $\text{Hom}_{\text{groups}}(\mathbb{Z}/n\mathbb{Z}, G)$ must also be iso to subgroup in G . For all $f \in \text{Hom}_{\text{groups}}(\mathbb{Z}/n\mathbb{Z}, G)$ there exists a element $g \in G$ s.t $f([1]_n) = g$ then $f([1]_n^n) = g^n$ but $[1]_n^n = [0]_n$ and since f is a homomorphism $f([0]_n) = e_G = g^n$. Thus $\text{Hom}_{\text{groups}}(\mathbb{Z}/n\mathbb{Z}, G) \cong \{g \mid g \in G : g^n = e_G\}$ or the set of elements of order n . $\square$

Isomorphisms

Problem 6.1. Let G' be the group of real matrices of the form $\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix}$. Is the map $\mathbb{R}^+ \rightarrow G'$ that sends x to this matrix an isomorphism?

Solution. Let $f : \mathbb{R}^+ \rightarrow G'$ take $\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \rightarrow x$ and let $g : G' \rightarrow \mathbb{R}^+$ take $a \rightarrow \begin{bmatrix} 1 & a \\ & 1 \end{bmatrix}$. $\forall A, B \in G'$ $AB = \begin{bmatrix} 1 & ab \\ & 1 \end{bmatrix}$. Then $f(AB) = ab = f(A)f(B)$. Thus f is a homomorphism. Furthermore $\forall A \in G'$ $(g \circ f)(A) = g(a) = A$ and for all $a \in G'$ $(f \circ g)(a) = f(A) = a$. Thus f and g are inverses and are bijective. Thus f is an isomorphism. ■

Problem 6.2. Describe all homomorphisms $\phi : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$. Determine which are injective, which are surjective, and which are isomorphisms.

Solution. Since $\mathbb{Z}^+$ is by definition a cyclic group whose generator is 1. Then for all $x \in \mathbb{Z}$, $\phi(x) = \phi(1^x) = \phi(1)^x$ (Note exponentiation is in context of groups, in this context it just means repeated addition). Thus $\phi(1)$ is the only value we care about for any homomorphism from $\mathbb{Z}^+ \rightarrow \mathbb{Z}^+$. So if $\phi(1) = n$ then $\phi(x) = nx$.

Thus all homomorphisms ϕ will take $x \rightarrow nx$ for $n \in \mathbb{Z}$ where $n = \phi(1)$. Let ϕ_n be a homomorphism s.t $\phi_n(1) = n$

All ϕ_n s.t $n \neq 0$ are injective. Because for all $a, b \in \mathbb{Z}^+$ if $\phi(a) = \phi(b)$ then $na = nb$ then $na - nb = 0$. Then $n(a - b) = 0$. We know $\phi(1) \neq 0$. Thus $a - b = 0$ and $a = b$. Thus for all ϕ_n where $n \neq 0$ are injective. If $n = 0$, then $n(a - b) = 0(a - b) = 0$ for all $a, b \in \mathbb{Z}^+$.

Only ϕ_1 and ϕ_{-1} . ϕ_1 is by definition the identity which is by definition surjective. For all $y \in \mathbb{Z}^+$, there exists $-y \in \mathbb{Z}^+$ s.t $\phi_{-1}(-y) = y$. Thus ϕ_{-1} is surjective.

Let ϕ_n be surjective. Then $\forall x \in \mathbb{Z}^+$, there exists a $y \in \mathbb{Z}^+$ s.t $\phi_n(y) = x = ny$. Thus $n|x$ for all $x \in \mathbb{Z}^+$. Suppose $x = 2$ or the smallest prime number this is only true when $n = \pm 1$. Thus $n = \pm 1$.

Thus only ϕ_1 and ϕ_{-1} are injective and surjective. ■

Problem 6.3. Show that the functions $f = 1/x$, $g = (x-1)/x$ generate a group of functions, the law of composition being composition of functions, that is isomorphic to the symmetric group S_3 .

Proof. $S_3 = \langle x, y \mid y^3 = x^2 = e, xy = y^2x \rangle$.

Let $G = \langle f, g \rangle$. The identity element of G is x . $f \circ f = \frac{1}{\frac{1}{x}} = x$. Thus order of f is 2. $g \circ g = \frac{-1}{x-1}$ and $g^2 \circ g = x$. Thus $|g| = 3$. Furthermore $f \circ g = \frac{x}{x-1}$ and $g^2 \circ f = \frac{-x}{1-x} = \frac{x}{1-x}$. Thus f is iso to x and g is iso to y and thus G is iso to S_3 because it follows its rules and generator structure. □

Claim 6.4. In a group, the products ab and ba are conjugate elements.

Proof.

$$\begin{aligned} a^{-1}(ab)a &= (a^{-1}a)ba \\ &= ba \end{aligned}$$

Thus there exists a element $g = a$ s.t $g^{-1}abg = ba$. Thus the products are conjugate elements. □

Problem 6.5. Decide whether or not the two matrices $A = \begin{bmatrix} 3 & \\ & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$ are conjugate elements of the general linear group $GL_2(\mathbb{R})$.

Proof. Let $A = \begin{bmatrix} 3 & \\ & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$. The characteristic polynomial of B is $P_B(x) = (1-x)(4-x) + 2 = x^2 - 5x + 6 = (x-3)(x-2)$. Thus the characteristic polynomial has distinct roots $x = 3, 2$. By a previous homework, we know that B is diagonalizable thus $B = P^{-1}DP$ and there exists a an eigenvector basis. Let P be the automorphism which takes the standard basis to the eigenvector basis s.t the eigenvector correspondding to 3 is first and eigenvector correspondding to 2 is second. Since we need to make sure $An_i = x_i n_i$ D must have 3 and 2 on the diagonal thus $D = B$. $\square$

Problem 6.6. Are the matrices $A = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ conjugate elements in $GL_2(\mathbb{R})$ and $SL_2(\mathbb{R})$

Proof. Yes, A and B are conjugate through $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ in $GL_2(\mathbb{R})$. For $SL_n(\mathbb{R})$, we look at the general matrix $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ which needs to satisfy the equation $AP = PB$ if A and B are conjugates. $AP = \begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix}$ and $BP = \begin{bmatrix} a+b & b \\ c+d & d \end{bmatrix}$. Thus $a+b = a+c$, $b+d = b$, $c+d = c$, and $d = d$. Thus $b = c$ and $d = 0$. Thus $P = \begin{bmatrix} a & b \\ b & 0 \end{bmatrix}$. $|P| = -b^2$ and if $P \in SL_2(\mathbb{R})$, then $-b^2 = 1$ and $b^2 = -1$ which isn't possible if $b \in \mathbb{R}$. Thus A and B are not conjugate in $SL_n(\mathbb{R})$. $\square$

Problem 6.10A. Find all automorphisms of a cyclic group of order 10

Solution. All cyclic groups of nth order are iso to $\mathbb{Z}/n\mathbb{Z}$ because you take the generator to $[1]_n$ and vice versa. Let G be any cyclic group of order 10 thus $G \cong \mathbb{Z}/10\mathbb{Z}$. Thus we are looking for the set $\text{Hom}_{\text{groups}}(\mathbb{Z}/10\mathbb{Z}, G) \cong \text{Hom}_{\text{groups}}(\mathbb{Z}/10\mathbb{Z}, \mathbb{Z}/10\mathbb{Z}) = B$. By problem 2C we know $\text{Hom}_{\text{groups}}(\mathbb{Z}/10\mathbb{Z}, G) \cong \{g \mid g \in G : g^{10} = e_G\} \cong \{[x]_{10} \mid [x]_{10} \in \mathbb{Z}/10\mathbb{Z} : [x]_{10}^{10} = [0]_{10}\} \cong B$. Thus we are looking for the set of $[x]_{10}$ whose order is 10. Thus the elements in the target set must generate $\mathbb{Z}/10\mathbb{Z}$. By the previous homework we know that only elements $[x]_n$ s.t x is coprime to 10 can generate $\mathbb{Z}/10\mathbb{Z}$. Thus only $[1]_{10}, [3]_{10}, [7]_{10}, [9]_{10}$ can generate $\mathbb{Z}/10\mathbb{Z}$. Define $f_x : \mathbb{Z}/10\mathbb{Z} \rightarrow \mathbb{Z}/10\mathbb{Z}$ which takes $[1]_{10} \rightarrow [x]_{10}$. Then the set of automorphisms is $\{f_x \mid \gcd(10, x) = 1\}$. $\blacksquare$

Problem 6.10B. Find all the automorphisms of S_3 .

Solution. The generator of $S_3 = \langle x, y \mid y^3 = x^2 = e, xy = y^2x \rangle$ is x and y . Thus every element in S_3 is a sequence of compositions of x and y . Furthermore any homomorphism, ϕ keeps those compositions intact and keeps the group structure intact so for every element since we can represent it as a sequence of x and y compositions and then act on the sequence and just act on x and y individually for any homomorphism on S_3 it only matters where x and y map to.

Assume the contrary, isomorphisms don't preserve the order of a element in a group. Let G be a group and $x \in G$ let $|x| = n$. Let $f : G \rightarrow H$ where H is also a group and f is a isomorphism. Then $f(x^n) \neq f(x)^n$. However $f(x * \dots * x) = f(x) * \dots * f(x) = f(x)^n$ thus forming a contradiction and isomorphisms keep order of a element.

And since automorphism are isomorphisms from a group to itself, thus there are at most 6 possibilities. Because x can only map to x or x^2 and y can only map to y, y^2, y^3 .

Claim 6.10BI. *There is a natural homo from $G \rightarrow Aut(G)$*

Define $f : G \rightarrow Aut(G)$ as the following *forall* $g \in G$ and for all $x \in G$ $f(g)(x) = g^{-1}xg$. For all $g, h \in G$ and for all $x \in G$ $f(gh) = (gh)^{-1}xgh = h^{-1}g^{-1}xgh = (f(h) \circ f(g))(x)$. Thus f is a homomorphism.

Let the kernel of f be K . For all $k \in K$, $f(k) = id$ and $\forall g \in G$, $f(kg) = id * f(g) = f(g) * id = f(gk)$, thus $gk = kg$ and k is commutative. Thus K is the center of G .

$|S_3| = 6 = |Aut(G)| = |f_*(S_3)|$ where $f : S_3 \rightarrow Aut(S_3)$ is the natural homo. Thus the only automorphisms are the conjugation automorphisms. ■